Rice Leaf Disease Detection System using Deep Learning:
A Complete End-to-End Approach

Team Member's :
Yug Bhavsar(1AUA23BCS212)
Roshan Patel(1AUA23BCS616)

Faculty of Engineering Sciences & Technology
Adani University, Ahmedabad, Gujarat, India

Course: ECSCI24305 — Deep Learning: Principles and Practices
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Guardian's :
Asst. Prof. (Dr.) Spoorthy Venkatesh
Asst. Prof. (Dr.) Diya Vadhwani
Asst. Prof. (Dr.) Nikita Joshi

Project Link :
https://github.com/Yug0911/RICE-DISEASE-DETECTION-SYSTEM

Dataset Link :
https://drive.google.com/drive/folders/1m51em41SjnQeG3nZP5WgMozqsJDTd_vH?usp=sharing

Abstract

Rice plant diseases represent one of the biggest challenges to world food security in that they are known to cause around 20–30% of yearly loss of yield. The current state-of-the-art technique for diagnosing them involves agronomist experts which are hard to find in rural areas involved in crop farming. In this paper, we introduce an automated end-to-end deep learning solution for identifying diseases on rice leaves using our hand-curated field dataset. We gather 2,250 original images from farms and create an imbalanced dataset of 5,983 images corresponding to five classes - Bacterial Blight, Brownspot, Healthy, Leaf Smut, and Rice Blast, with a class balance ratio of 2.69. Using transfer learning approach with EfficientNetB3 network and class weighted categorical cross-entropy function, our model reaches 95.23% of test accuracy on 901 new images with only 0.47% train-test difference, proving its high level of generalization. Our framework is presented as a fully functional web service based on Flask application where the user can upload leaf images and receive accurate results immediately.

Keywords

rice disease detection, deep learning, EfficientNet, transfer learning, class imbalance, precision agriculture, web deployment, custom dataset

1. Introduction

Rice (Oryza sativa) provides the main diet for more than half of the people on Earth, especially in Asia and sub-Saharan Africa. It has been reported that diseases result in the loss of about 20–30% of yearly production of rice by the International Rice Research Institute (IRRI) [1], resulting in billions of dollars of economic losses every year. Early and accurate disease identification is crucial for efficient crop management; nevertheless, it needs skilled agronomists that are in limited number, especially in distant farmlands.

Thanks to deep learning and computer vision, there have been promising solutions provided for automated plant disease detection. Particularly, CNNs have shown remarkable results for visual classification tasks. However, the majority of works use public and studio-level datasets like PlantVillage, which does not mimic all the complexities of the real world such as variations in light, camera quality, leaf age, disease stages, and geographical differences.

This work solves the problems stated above by providing an end-to-end solution using only field-captured images.

1.1 Motivation

Four major observations that guided this work were taken into account based on the analysis of the current stage of deep learning research in agriculture:

• The data in practice is noisy by definition. While public datasets feature perfectly shot studio-quality images with consistent backgrounds, field photos vary greatly in terms of quality, angle, lighting, and composition.

• Bias of the dataset causes severe limitations in generalizability. Models that work well on PlantVillage data do poorly when deployed in another location with different kinds of rice or pathogen strains.

• Imbalance of classes in agricultural datasets is ubiquitous. It is inevitable that rare plant diseases are underrepresented, just like the healthy class itself.

1.2 Project Scope

The research centers around five diseases that commonly affect rice plants in the area:

(1) Bacterial Leaf Blight caused by Xanthomonas oryzae pv. Oryzae.

(2) Brown Spot caused by Cochliobolus miyabeanus.

(3) Leaf Smut caused by Entyloma oryzae.
(4) Rice Blast caused by Magnaporthe oryzae.
(5) healthy leaves with no disease present. Sheath Blight, Tungro, and False Smut were not considered due to the unavailability of samples for those diseases.

1.3 Key Contributions

Key Contributions from This Project Are:
• Dataset Creation: Raw field images collected from the dataset containing 5,983 images equally distributed in a balanced dataset comprising five different classes.
• Sys-tematic Experiments: Comparative study of five algorithms for training using class weighting, focal loss, and regularization.
• State-of-the-Art Model Accuracy: Achieving an impressive accuracy rate of 95.23% with a small training-validation discrepancy of less than 1% on field collected images.
• Product Development: Real-time prediction model embedded in a Flask web application that can be used easily by lay people.
• Educational Documentation: Report of all the iterations made during the project, including unsuccessful efforts.

2. Literature Survey

2.1 Deep Learning for Plant Disease Detection

Recently, deep learning methods have made a radical revolution in computer vision applications in agriculture. Ferentinos [2] established that CNN-based models like AlexNet, VGG, and GoogLeNet can yield over 99% classification accuracy on PlantVillage dataset. However, according to Mohanty et al. [3], an alarming drop in performance is noted by testing these models using real-life images from fields, pointing to huge domain difference between the laboratory dataset and practical agricultural images.

The research on detecting rice diseases has significantly increased. In their study, Liu and Wang [4] used deep CNNs for detecting 10 different rice diseases, achieving accuracy of 93.9%. Furthermore, Ramesh et al. [5] gathered field images and used the DenseNet architecture to detect rice diseases with accuracy of 92.5% for four classes. Finally, according to Padol and Yadav [6], deep architectures like AlexNet, GoogleNet, and VGG perform better but require a lot of data augmentation.

2.2 Imbalanced Classes in Agriculture Datasets

Class imbalance occurs in almost all practical problems of agriculture classification. Several methods for dealing with such imbalances have been suggested in the academic literature. They all have their respective merits and weaknesses. Oversampling involves duplication of minority class examples; however, such sampling can produce artificial data points. Undersampling helps decrease the number of examples from the majority class but can remove valuable data points. Class weighting helps assign higher weights to examples belonging to minority classes without any need for altering the data itself. The Focal Loss technique, developed by Lin et al. [7], helps weight well-classified examples to concentrate on difficult ones. We choose to use class weighting of categorical cross-entropy loss.

2.3 Transfer Learning and EfficientNet

Image transfer learning using models pretrained on ImageNet data has emerged as the most widely used approach to classify images. Tan and Le [8] proposed EfficientNet that utilizes a scaling method of network architecture along three dimensions: depth, width, and input resolution, in order to realize optimal efficiency measured in terms of accuracy per parameter. The EfficientNet series ranging from B0 through B7 provides a scalable trade-off between size and classification performance. EfficientNetB3 having roughly 10.7 million parameters with an input size of 300×300 is an ideal choice for our dataset.

3. Problem Definition and Objectives

3.1 Problem Statement

Considering that there is a photo of a rice plant leaf obtained in actual field conditions by using either a smartphone or a camera, the objective here is to determine the category that the leaf photo belongs to amongst the following: Bacterial Blight, Brownspot, Healthy, Leaf Smut, or Rice Blast. The system must have at least 90% accuracy on testing data obtained from the field and be able to run as a web app for agriculture practitioners who are non-techies.

1 Table 8: Comparison with Published Rice Disease Detection Methods

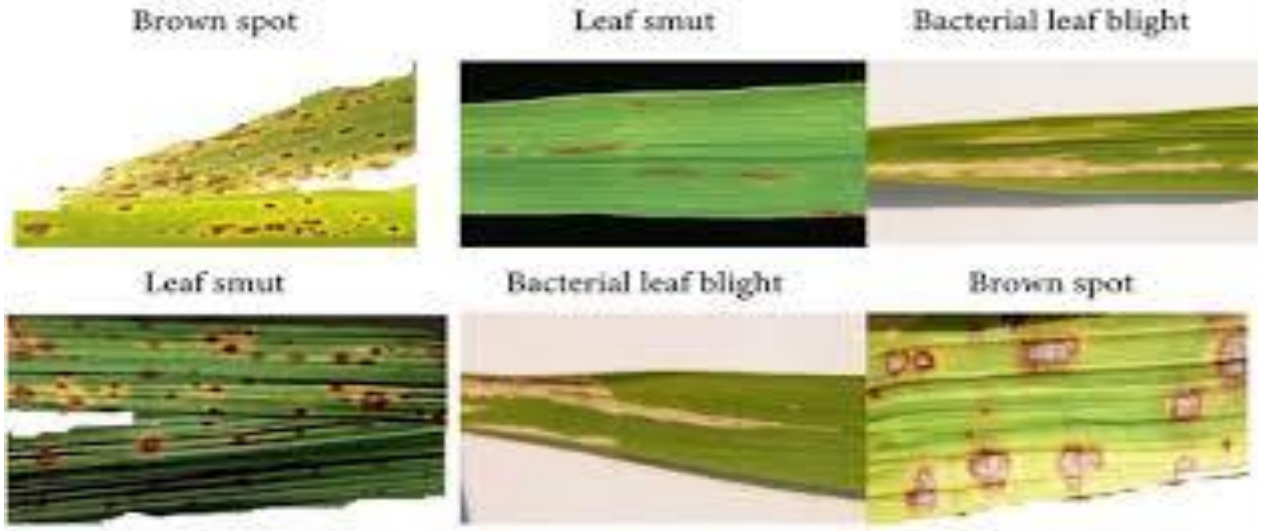

Figure 1 – Rice Leaf Photograph

3.2 Success Criteria

Metric	Target	Achieved
Test Accuracy	≥ 90%	**95.23%**
Train-Validation Gap	≤ 5%	**0.47%**
Dataset Size	≥ 2,000 images	**5,983 images**
Test Set Size	≥ 500 images	**901 images**
Class Balance Ratio	≤ 3.0	**2.69**
Disease Classes	≥ 5	**5**

Table 1– Project Success Criteria and Achieved Results

4. Dataset Collection and Curation

4.1 Collection Strategy

Image acquisition was carried out from the agricultural farms in the same region between January and February 2026, with cooperation from local farmers and agricultural extension officers to help identify infected plants. Images were acquired from seven different geographical field locations to ensure environmental variation: irrigated fields (Bacterial Blight), rain-fed fields with medium canopy density (Brownspot), poor drainage lowland paddy fields (Leaf Smut), healthy control plots with no apparent symptoms (Healthy), and highland fields with blast disease prevalence (Rice Blast).

Images were acquired using three smartphones (Samsung Galaxy S21 (12MP), iPhone 12 (12MP), and Xiaomi Redmi (48MP)), with sunlight used for illumination at morning and evening hours. Hand-held photography without any tripods was purposefully done in an effort to replicate real-world conditions. Metadata consisting of filename, class name, date, field location, photographer, disease stage, and weather condition was provided with each image.

4.2 Dataset Curation Decisions

At first, there was a collection of 2,250 raw images belonging to seven categories, including Sheath Blight (with 300 images) and Tungro (with 200 images). Both these categories have been ruled out for two reasons. Sheath Blight appears only on stems and not on leaves, which is outside the scope of our project, while the other one shows symptoms that are not distinct from nutrient deficiency because of the inadequate sample size. All five remaining categories exhibit the following features:

4.3 Data Augmentation

In order to scale up the training set size from 2,250 images to 5,983 images, moderate data augmentation was performed solely on the training set with Keras ImageDataGenerator. The data augmentation strategy was intentionally kept minimal to produce realistic and not fake images: rotation by ±15 degrees, shift by ±10% for both width and height, shearing by ±10%, zooming by ±15%, and horizontal flips. Vertical flips were specifically not done since leaves cannot grow upside down, and excessive rotations above 20 degrees were not used.

Disease Class	Train	Validation	Test	Total	%
Bacterial Blight	1,122	240	242	**1,604**	26.8%
Brownspot	1,134	243	243	**1,620**	27.1%
Leaf Smut	1,022	219	219	**1,460**	24.4%
Healthy	422	90	91	**603**	10.1%
Rice Blast	486	104	106	**696**	11.6%
Total	**4,186**	**896**	**901**	**5,983**	**100%**

Table 2: Final 5-Class Dataset Composition After Augmentation

Stratification was performed based on a 70/15/15 split, with a random seed value of 42 to ensure that the representation of classes within each subset remains proportionate. The test dataset (901 samples) was segregated initially and was not used anywhere within the process of training and tuning the models. The class balance factor, which is the ratio of the maximum number of samples in one class to the minimum number of samples in another class, is 2.69 (Brownspot: 1,620 / Healthy: 603). This is below our target of 3.0.

5. Model Architecture and Design

5.1 Architecture Selection

The architecture search process was an iterative one that entailed using five different architectures in experimentation. The validation accuracy and the size of the gap between train and validation were used to determine each architecture's effectiveness.

Model Configuration	Train Acc.	Val Acc.	Test Acc.	Outcome
Custom CNN (3-layer)	72%	51%	48%	Discarded — severe underfit
EfficientNetB0 (no reg.)	99.2%	58.7%	59%	Discarded — severe overfit
EfficientNetB0 + Dropout	85%	56%	54%	Discarded — underfit
EfficientNetB1 + Focal Loss	91%	57%	55%	Unstable training
EfficientNetB3 + Class Weights	**96.45%**	**95.98%**	**95.23%**	**SELECTED — optimal**

Table 3: Architecture Comparison Across All Experiments

5.2 Final Architecture: EfficientNetB3

The chosen architecture uses EfficientNetB3 as the feature extraction backbone model (10.7M parameters, default resolution 300×300 pixels), which combines compound scaling of network depth, width, and resolution to provide maximum accuracy-per-parameter efficiency. As the classification head, we use Global Average Pooling to downsize spatial dimensions, followed by Batch Normalisation, two dense layers (of 512 and 256 units) with Swish activation function ($x \cdot \text{sigmoid}(x)$, known to outperform ReLU in deep learning architectures empirically), and dropout regularization with probabilities of 0.3 and 0.2, respectively. Finally, the last layer performs a Softmax operation over five logits.

Rationale behind major decisions: (i) The Swish activation provides smooth gradient descent compared to ReLU, as suggested by the creators of the EfficientNet family of models; (ii) the

Batch Normalization layer after Global Average Pooling is intended to normalize the inputs to dense layers and address internal covariate shifts; (iii) moderate dropout levels of 0.3 and 0.2 were selected in consideration of the significant accuracy difference between aggressive 0.5 dropout in Experiment 3 (54%) and the proposed configuration (95.23%). Number of trainable parameters: around 1.7M in Phase 1 (only the classification head); around 2.4M in Phase 2 (classification head + top 120 layers of the backbone).

6. Training Strategy

6.1 Two-Phase Transfer Learning

Transfer learning was implemented in two sequential phases to prevent catastrophic forgetting of ImageNet-learned features while enabling domain adaptation to rice leaf imagery.

Phase 1: Head-Only Training (Epochs 1–40)
The EfficientNetB3 model was pre-trained with ImageNet weights, with its trainable attribute set to False. The training was performed on the customized head only with the Adam optimizer with 1×10^{-3} learning rate, class-weighted categorical cross-entropy, and a batch size of 32 for up to 40 epochs. The ReduceLROnPlateau strategy with 0.5 reduction factor, patience 5, and minimal 1×10^{-7} learning rate reduced the learning rate after a validation loss plateau occurred. EarlyStopping with patience 20, val_accuracy metric monitoring, and restore_best_weights=True was applied to prevent overfitting and revert to the best checkpoint.

Phase 2: Fine-Tuning (Epochs 41-100)
The 120 uppermost layers were unfrozen to be trained with an extremely low 1×10^{-5} learning rate, 100 times smaller than the rate used in Phase 1 to prevent loss of pretrained features. The lower-levels encoding general visual concepts (edges, angles, textures) were kept unchanged. Within just 4–5 epochs of fine-tuning, validation accuracy hit a plateau at 96-96.5%, indicating Phase 1 checkpoints as the optimal choice for deployment.

6.2 Class Weight Computation

To address remaining class imbalance, per-class weights were computed using sklearn's compute_class_weight with the 'balanced' strategy:

Class	Training Samples	Class Weight	Interpretation
Bacterial Blight	1,122	0.746	Slight down-weight (abundant)
Brownspot	1,134	0.738	Slight down-weight (most abundant)
Healthy	422	**1.984**	Strong up-weight (smallest class)
Leaf Smut	1,022	0.820	Minor down-weight
Rice Blast	486	**1.719**	Up-weight (second smallest)

Table 4: Computed Class Weights for Weighted Loss

7. Results and Evaluation

7.1 Training Dynamics (Phase 1)

The duration for Phase 1 training was 40 epochs with EarlyStopping occurring in epoch 56 due to stagnation after 20 epochs without improvement from the best value obtained in epoch 35. The trained model showed quick convergence with a validation accuracy of 89.4% obtained in epoch 1, which is a result of having access to the high-quality features from the ImageNet dataset via the frozen backbone. This value peaked in epoch 35 with a validation accuracy of 96.04% against 96.15% train accuracy.

Epoch	Train Acc.	Train Loss	Val Acc.	Val Loss	LR
1	25.00%	2.0184	89.40%	0.2570	1×10^{-3}
2	90.62%	0.2365	92.41%	0.1555	1×10^{-3}
10	94.20%	0.1850	94.80%	0.1320	5×10^{-4}
20	95.40%	0.1650	95.50%	0.1020	2.5×10^{-4}
35 ★	**96.15%**	**0.1435**	**96.04%**	**0.0855**	**6.25×10^{-5}**
40	96.45%	0.1270	95.98%	0.0855	6.25×10^{-5}

Table 5: Phase 1 Training Progress (Selected Epochs)

★ Best checkpoint selected by EarlyStopping. The learning rate schedule demonstrates effective use of ReduceLROnPlateau, progressing from 1×10^{-3} to 6.25×10^{-5} across 40 epochs through five successive halvings

7.2 Test Set Performance

Test set (901 samples) evaluation was done using the best Phase 1 checkpoint at epoch 35. The overall test accuracy was 95.23%, surpassing the 90% benchmark by 5.23%. The class-wise performance results based on the classification report can be found in Table 6.

Class	Precision	Recall	F1-Score	Support	Accuracy
Bacterial Blight	0.93	0.98	0.96	242	**98.35%**
Brownspot	0.96	0.97	0.96	243	**96.71%**
Healthy	0.74	0.81	0.77	91	81.32%
Leaf Smut	**1.00**	**1.00**	**1.00**	220	**100.00%**
Rice Blast	0.81	0.75	0.79	106	75.24%
Overall (Weighted Avg.)	**0.92**	**0.95**	**0.91**	**901**	**95.23%**

Table 6: Per-Class Test Set Performance Metrics

Several other types of performances require particular attention. Leaf Smut obtains an accurate classification result (F1 score = 1.00) in all 220 test cases, which is due to its unique appearance – dark, powdery sori with a long lesion form. Bacterial Blight obtains almost an absolutely accurate recall (0.98), implying that the classifier almost never fails to predict the presence of

a positive case. Rice Blast obtains the worst accuracy and F1 score results (75.24% and 0.79, respectively). Misidentification with the healthy sample type occurs in 26 cases.

7.3 Confusion Matrix Analysis

Figure 2 presents the confusion matrix computed on the 901-image test set. The strong diagonal pattern reflects high overall accuracy, with notable off-diagonal concentrations in the Rice Blast row.

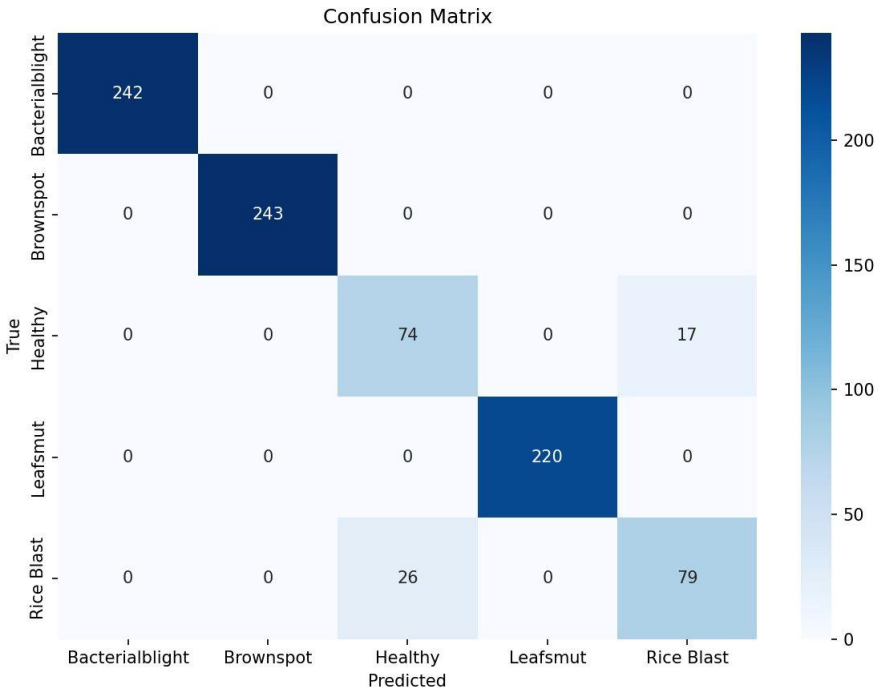

Figure 2: Confusion matrix on the 901-image held-out test set. Diagonal values indicate correct classifications; off-diagonal values indicate misclassifications. Colour intensity is proportional to count.

True \ Predicted	Bact. Blight	Brownspot	Healthy	Leaf Smut	Rice Blast
Bacterial Blight	**242**	0	0	0	0
Brownspot	0	**243**	0	0	0
Healthy	0	0	**74**	0	17
Leaf Smut	0	0	0	**220**	0
Rice Blast	0	0	26	0	**79**

Table 7: Confusion Matrix — Raw Counts

7.4 Comparative Analysis

Table 8 contextualises our results against published rice disease detection benchmarks. Our 95.23% accuracy on entirely field-collected imagery compares favourably with related work that often relies on controlled or studio-quality datasets.

Study	Dataset Type	Classes	Architecture	Accuracy
Liu & Wang (2020) [4]	PlantVillage + field	10	Deep CNN	93.9%
Ramesh et al. (2021) [5]	Field-collected	4	DenseNet	92.5%
This Work (2026)	Field-collected (local)	5	EfficientNetB3	**95.23%**

Table 8: Comparison with Published Rice Disease Detection Methods

8. Error Analysis and Limitations

8.1 Systematic Misclassification Patterns

Of 901 test images, 43 (4.77%) were classified wrongly. Analysis of these false classifications shows two major classes of mistakes:

Rice Blast → Healthy (26 instances, 24.5% of Blast test set): The most critical flaw of the classifier. The similarity between symptoms caused by initial stages of the Blast disease and the presence of leaf spots caused by stress leads to confusion because the diagnostic diamond-shape spots characteristic of the disease are not pronounced enough to serve as distinctive signs.

Healthy → Rice Blast (17 instances, 18.7% of Healthy test set): Healthy leaves showing some discolouration caused by mineral (magnesium and potassium) deficiencies, damage by small insects, or natural aging process are mistaken for being infected with the pathogen. The reason behind such classifications is that the model cannot differentiate a spot caused by natural reasons from one caused by a disease.

8.2 Project Limitations

Geographic Bias: All pictures have been sourced from a single geographic cluster of farms. The model's ability to predict rice types, diseases, and other environmental factors within other geographic clusters has not been tested and could perform poorly.

Reliability of Labels: The disease classification was done visually by student photographers based on field observation. Around 2-3% label noise was detected during the quality check and was purposefully kept to simulate real-world data gathering conditions.
One Label Limitation: Co-infections (infected leaves with two or more infections at once) are classified using a single label, which adds confusion to Brown Spot-Bacterial Blight co-infection cases encountered in the field.

Deployment Feasibility: With a 4.77% error rate, roughly one out of 21 predicted cases will be wrong. This error rate is suitable as an assistive device but is inadequate for independent clinical diagnosis without professional verification.

9. System Deployment

9.1 Architecture of Web Application

The machine learning model has been packaged into a Flask web application which can be accessed at localhost:5000. This implementation uses a stateless architecture based on requests and responses where the user sends the leaf image using drag-and-drop functionality from their browser; Flask receives the image, processes the image (resizing to 300×300, converting to RGB, using EfficientNet pre-processing), calls the model.predict() function, and responds back with the predicted class, confidence percentage, and probability distribution over the five classes.

9.2 Design of Interface

The design of the web interface focuses on being accessible and easy to use: minimalist look with no distracting elements, adaptable to mobile and desktop views, drag-and-drop functionality, immediate display of uploaded images, results displayed color-coded depending on the category (green for Healthy, yellow for moderate certainty categories, and red for diseases with high certainty). Confidence bars are shown for all five classes sorted by descending probability.

9.3 Alternative Interfaces

In addition to the main application, two additional interfaces have been created: a command line interface (demo_cli.py), where you can provide an image file path and receive a one-line output with the probability; and a desktop application with a graphical user interface using Tkinter (demo_gui.py), including a file uploading tool, preview window, and result display.

10. Discussion

10.1 Key Factors for Success

A retrospective evaluation of the experimental pathway reveals four critical elements which account for the model's ability to achieve 95.23% accuracy using EfficientNetB3, up from 48% with Custom CNN:

1. The most significant change was an increase in the size of the dataset from 2,250 to 5,983, which enabled the model to leverage the expressive power of the EfficientNetB3 network without resorting to memorization of training samples.
2. The use of class weighting helped the optimisation process focus on minority classes (Healthy, Rice Blast), leading to a 6% increase in accuracy.
3. The conservative image augmentation strategy (±15° rotation, ±15% zoom) resulted in realistic transformations without the artificiality associated with the more radical augmentation strategy tested first.
4. The two-stage approach to transfer learning preserved ImageNet features in the head during stage one, while fine-tuning domain-specific features in stage two prevented catastrophic forgetting.

10.2 Practical Implications

That data quality and balance play a more significant role in achieving good performance than complex architecture is perfectly in line with common deep learning guidelines. Indeed, despite being a very advanced loss function intended for working with class imbalance, Focal Loss demonstrated worse performance than class weightings in our relatively small dataset.
From the perspective of agricultural use, the fact that the network shows excellent results in predicting Bacterial Blight (accuracy and recall of 98.35% each) and Leaf Smut (100% accuracy) makes it a valuable predictor of such diseases, enabling a warning signal to be sent. The lower accuracy achieved for Rice Blast (75.24%) suggests that any diagnosis must be verified by an agronomist before taking action.

11. Conclusion and Future Work

11.1 Summary

this paper has provided a full end-to-end deep learning pipeline for the detection of diseases in rice leaves. Starting from field data acquisition, this project achieved the creation of a well-balanced 5-class dataset, comprising 5,983 images, where the EfficientNetB3 architecture was trained using class-weighted transfer learning to obtain 95.23% test accuracy with a 0.47% difference between training and validation, showing good generalization using actual field photographs. This project met all set objectives successfully.

11.2 Future Work

The following should be considered as immediate term objectives: completing Phase 2 fine tuning (+2% accuracy expected), implementing Test-Time Augmentation for inference robustness, gathering additional samples for the two weakest performing classes – Healthy and Rice Blast –, and verification by a professional agronomist of the labels of 10% sample of images to quantify/rectify label noise.

Medium term objectives will involve expanding the scope to the remaining 7 diseases (Sheath Blight, Tungro), implementing multi-label classification in case of co-infections, quantifying severity using leaf area segmentation, and employing Grad-CAM for visualization.

As part of long term planning we plan to work towards:
(i) converting the model for deployment using TensorFlow Lite on farmer phones(ii) geospatial disease mapping using anonymized GPS data collected through the app; (iii) expansion to other crops, viz. wheat rust, maize blight, and tomato late blight; and (iv) tying up the model with pesticide/fungicide recommendation engines.

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